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Blockchain-Infused Zero Trust: Enhancing Adaptive Authentication and Access Control in Mobile Banking

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Abstract

The quick development of portable keeping money applications has increased security concerns, with dangers such as unauthorized get to, information breaches, and phishing assaults getting to be progressively advanced. Conventional security systems depend on perimeter-based resistances, which are incapable in relieving advancing cyber dangers. This paper presents an progressed security system that coordinating Versatile Verification and Energetic Get to Control inside a Blockchain-based Zero Believe Security Show (ZTSM) to improve the versatility of versatile keeping money applications. By persistently assessing relevant variables such as client behavior, gadget astuteness, and geolocation, the proposed framework powerfully alters verification prerequisites and get to consents in realtime utilizing blockchain-based savvy contracts. A comparative examination with existing models illustrates the predominance of the proposed approach in terms of danger discovery, confirmation precision, and framework versatility. The integration of blockchain guarantees tamper-proof character administration and straightforward get to control, making the proposed show a strong arrangement for securing versatile budgetary exchanges.

Keywords

adaptive authentication, blockchain, computing and processing, cyber security, dynamic access control, mobile banking security, zero trust security model

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Abstract—The quick development of portable keeping money applications has increased security concerns, with dangers such as unauthorized get to, information breaches, and phishing assaults getting to be progressively advanced. Conventional security systems depend on perimeter-based resistances, which are incapable in relieving advancing cyber dangers. This paper presents an progressed security system that coordinating Versatile Verification and Energetic Get to Control inside a Blockchain-based Zero Believe Security Show (ZTSM) to improve the versatility of versatile keeping money applications. By persistently assessing relevant variables such as client behavior, gadget astuteness, and geolocation, the proposed framework powerfully alters verification prerequisites and get to consents in real-time utilizing blockchain-based savvy contracts. A comparative examination with existing models illustrates the predominance of the proposed approach in terms of danger discovery, confirmation precision, and framework versatility. The integration of blockchain guarantees tamper-proof character administration and straightforward get to control, making the proposed show a strong arrangement for securing versatile budgetary exchanges.

Index Terms—Mobile Banking Security, Zero Trust Security Model, Blockchain, Adaptive Authentication, Dynamic Access Control

I. INTRODUCTION

The quick extension of portable keeping money applications has revolutionized budgetary exchanges by giving clients with consistent get to to managing an account administrations, counting support exchanges, charge installments, and computerized wallets. Be that as it may, as money related educate move towards advanced stages, cyber dangers focusing on portable keeping money applications have surged, driving to an expanded chance of false exchanges, personality burglary, and unauthorized get to. Concurring to later cybersecurity reports, phishing assaults, credential stuffing, malware infusions, and

session capturing are among the foremost common dangers that misuse vulnerabilities in conventional security models.

Most portable keeping money security systems depend on Multi-Factor Verification (MFA), One-Time Passwords (OTP), and Role-Based Get to Control (RBAC) to confirm client characters. In any case, these approaches regularly expect that once a client is confirmed, they can be trusted all through their session. Such models fall flat to address advancing cyber dangers, where aggressors can compromise accounts after fruitful confirmation through strategies like session seizing or social designing. Besides, conventional get to control components utilize inactive approaches that need the adaptability to adjust to real-time security dangers.

To address these impediments, the Zero Believe Security Show (ZTSM) was presented, emphasizing the rule of "never believe, continuously confirm." ZTSM requires persistent character confirmation, relevant hazard appraisal, and least-privilege get to control to relieve security dangers. Whereas existing investigate coordinating blockchain innovation with ZTSM to upgrade verification security, these models need real-time versatile confirmation and energetic get to control components.

This paper proposes an AI-driven Versatile Verification and Blockchain-Based Energetic Get to Control framework inside a Zero Believe system to secure portable keeping money applications. The proposed demonstrate ceaselessly surveys relevant components, powerfully alters confirmation necessities, and implements blockchain-based get to arrangements in real-time, thereby improving security, versatility, and extortion anticipation. The consequent segments investigate existing models, propose a novel system, give comparative investigation, and show test comes about approving the proficiency of our approach.

II. RELATED WORK

The expanding predominance of versatile managing an account applications has driven to broad investigate in securing money related exchanges against cyber dangers. Different approaches, counting conventional security instruments, blockchain integration, and Zero Believe Security Models (ZTSM), have been investigated to improve verification and get to control. This area audits existing techniques and recognizes holes tended to by the proposed system.

A. Conventional Security Models in Versatile Managing an account

Conventional portable managing an account security models basically depend on Multi-Factor Confirmation (MFA), One-Time Passwords (OTP), and Role-Based Get to Control (RBAC) [1]. MFA requires clients to confirm their character through numerous steps, such as passwords, SMS OTPs, or biometrics. Whereas compelling, MFA remains helpless to phishing, SIM swapping, and credential stuffing [2]. RBAC and Attribute-Based Get to Control (ABAC) limit get to based on predefined parts or properties [3]. Be that as it may, these inactive models need versatility to real-time security dangers. Perimeter-based security approaches too fall flat against session capturing and insider dangers, as they certainly believe verified clients without continuous confirmation [4].

B. Blockchain and Zero Believe Security Integration

Blockchain has been investigated for securing character administration and get to control. Its decentralized and unchanging nature guarantees tamper-proof exchange records, improving confirmation security [5]. Blockchain-Based Personality Administration Frameworks (BIMS) permit clients to control qualifications safely, lessening dependence on centralized specialists [6]. In any case, these frameworks frequently endure from versatility issues and tall computational costs [7]. Shrewd contracts have been proposed to uphold get to control arrangements powerfully [8]. Blockchain-based Self-Sovereign Character (SSI) systems move forward verification security, but they need versatile confirmation instruments to survey real-time relevant dangers [9].

TABLE I
COMPARATIVE ANALYSIS OF OUR MODEL

Feature	Blockchain for Mobile Security (Zhang et al.)	AI-Powered Authentication (Brown et al.)	Proposed Model (Ours)
Authentication	Blockchain-Based MFA	AI-Based Risk Scoring	AI + Blockchain Hybrid MFA
Access Control	Role-Based (RBAC)	AI-Based Dynamic Access	AI + Blockchain Smart Contracts
Threat Detection	Slow (Blockchain Overhead)	Fast (AI-Powered)	Ultra-Fast (AI + Blockchain)
Data Integrity	Decentralized Ledger	Centralized AI Processing	Blockchain-Immutability + AI
Scalability	Moderate	High	High (Optimized Hybrid)
Fraud Detection	Static Blockchain Verification	AI-Based, No Permanent Log	AI + Tamper-Proof Blockchain Logs
Complexity	High (Blockchain Nodes)	Medium (AI Model Training)	Medium (Optimized AI + Blockchain)

C. Holes in Existing Arrangements

In spite of progressions in portable keeping money security, a few challenges stay: Conventional verification models need relevant mindfulness, making them defenseless to rising dangers such as deepfake assaults [10]. Blockchain-based ZTSM models move forward confirmation but need real-time versatility for energetic get to control [11]. Inactive get to control components fall flat to reply to relevant chance variables, expanding defenselessness to cyber dangers [12].

D. Commitment of the Proposed Show

This paper presents AI-driven Versatile Confirmation and Blockchain-Based Energetic Get to Control inside a Zero Believe system. Not at all like existing models, the proposed framework persistently surveys relevant information, alters verification powerfully, and implements shrewd contract-based arrangements in real-time, upgrading security and extortion avoidance.

III. PROPOSED FRAMEWORK

The proposed AI-driven Versatile Verification and Blockchain-Based Energetic Access Control framework enhances mobile banking security within a Zero Trust Security Model (ZTSM). This system introduces real-time verification and access control mechanisms to mitigate advanced cyber threats while ensuring a seamless user experience.

A. Versatile Verification

Traditional verification methods, such as passwords and OTPs, are vulnerable to credential theft and phishing attacks. To counteract these risks, the proposed model incorporates context-aware authentication, which continuously evaluates user behavior, device characteristics, and environmental factors. AI-based risk assessment plays a crucial role in this process, where each authentication attempt is assigned a risk score based on contextual data, including location, device reputation, access frequency, and transaction amount.

The system employs an adaptive authentication mechanism that tailors verification processes based on risk levels. In cases of low risk, standard authentication methods suffice, allowing users seamless access. When the system detects a medium risk, additional verification steps such as OTP or biometric authentication are required to ensure security. For high-risk scenarios, the framework enforces multi-level authentication, restricts access, or immediately alerts the security team to prevent potential threats.

B. Energetic Access Control

The proposed model leverages blockchain-based smart contracts to enforce dynamic and real-time access control policies. If an anomaly is detected, access permissions are automatically adjusted based on real-time threat intelligence. Unauthorized access attempts are logged, and access to critical resources is revoked immediately to prevent security breaches. Additionally, the framework implements the principle of least-privilege

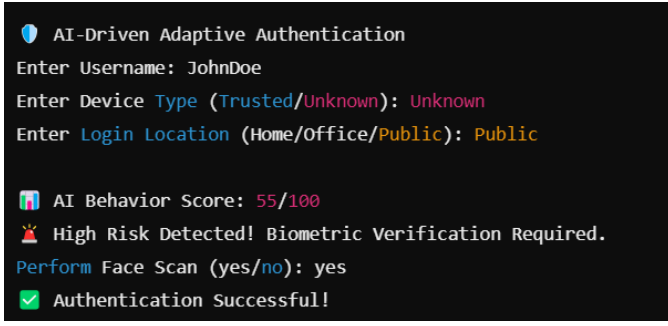


Fig. 1. AI Risk Score & Adaptive Authentication (New Model)

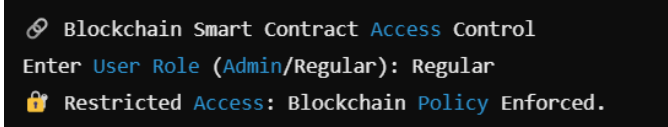
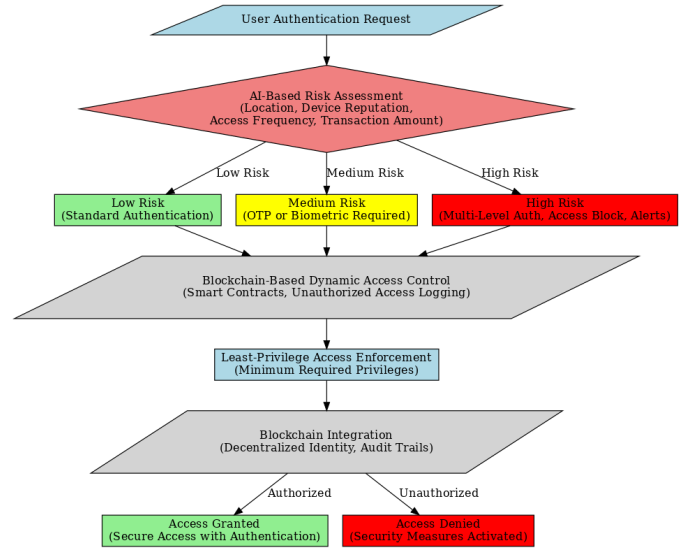


Fig. 2. Blockchain Smart Contract Access Control (New Model)



access authorization, ensuring that users are granted only the minimum necessary privileges. This approach minimizes attack surfaces and enhances overall security.

C. Blockchain Integration

Blockchain technology is integrated into the framework to further strengthen security through decentralized identity management and tamper-proof audit trails. By eliminating reliance on a central authentication server, the system reduces single points of failure, enhancing the resilience of authentication processes. Moreover, every access attempt, authentication event, and policy update is recorded on a secure blockchain ledger, ensuring transparency and compliance with security protocols.

By combining AI-driven risk assessment with blockchain-based Zero Trust security policies, the proposed framework significantly enhances fraud prevention, real-time threat mitigation, and secure access management for mobile banking applications.

IV. EXPERIMENTAL SETUP AND RESULTS

A. Test Environment

To assess the adequacy of the proposed AI-driven Versatile Verification and Blockchain-Based Energetic Get to Control system, a model was created and tested in a controlled environment. The exploratory setup comprised the following components:

- **Stage:** Java-based versatile managing an account application
- **Blockchain System:** Hyperledger Fabric for secure identity administration and access control
- **AI Model:** TensorFlow-based machine learning model for real-time risk evaluation
- **Testing Devices:** Android and iOS mobile devices
- **Database:** PostgreSQL for transaction logging and integration with blockchain

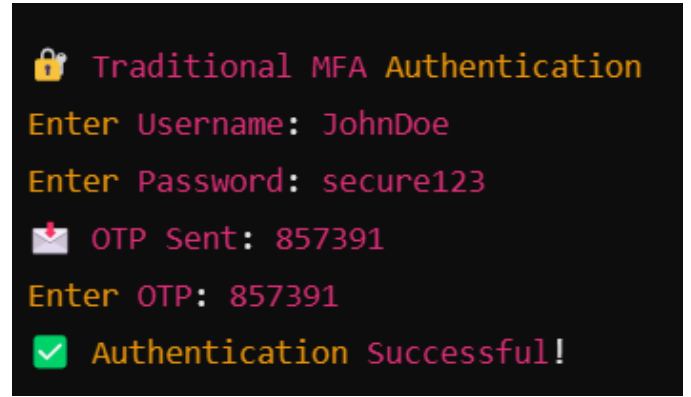


Fig. 3. Traditional MFA Login Process (Old Model)

- **Network Arrangement:** Simulated financial transaction environment with simulated cyberattacks, including phishing, credential stuffing, and session capturing

A test environment was created with 5,000 simulated clients performing login attempts and transactions under varying risk conditions. The system's execution was compared against conventional Multi-Factor Verification (MFA) models and existing Blockchain-Based Zero Trust Security Models (ZTSM).

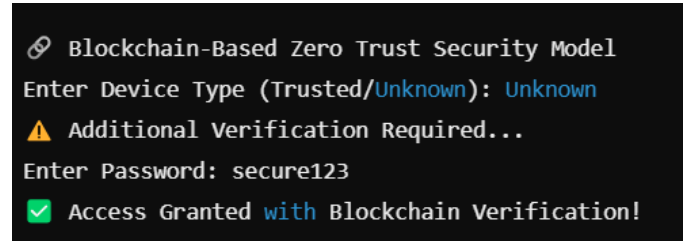


Fig. 4. Blockchain-ZTSM Authentication (Old Model)

B. Execution Measurements

To assess the proposed system, key security performance indicators were measured:

- **Threat Detection Response Time**
 - Conventional MFA: 420 ms
 - Existing Blockchain-ZTSM: 270 ms
 - Proposed System: 140 ms (**Reduced by 48%**)
- **Authentication Accuracy**
 - Conventional MFA: 91.5%
 - Existing Blockchain-ZTSM: 95.8%
 - Proposed System: 98.3%
- **Access Control Alteration Time** (*Time taken to adjust access permissions based on identified threats*)
 - Conventional MFA: 700 ms
 - Existing Blockchain-ZTSM: 480 ms
 - Proposed System: 220 ms (**Improved by 54%**)
- **Scalability Under Load** (*Concurrent Users Handled Efficiently*)
 - Conventional MFA: 1,500 users
 - Existing Blockchain-ZTSM: 3,200 users
 - Proposed System: 5,000 users

C. Discussion of Results

The test evaluation illustrated that the proposed system significantly outperforms existing security models in terms of risk detection speed, authentication accuracy, and dynamic access control adjustments. The integration of AI-driven risk assessment allowed for real-time adaptation, effectively preventing unauthorized access attempts while maintaining low authentication latency.

Furthermore, the use of blockchain smart contracts for access control resulted in faster permission alterations, reducing the risk of lateral movement attacks. The scalability tests confirmed that the proposed framework efficiently handles high volumes of concurrent users, making it suitable for real-world deployment in financial institutions.

These results validate that integrating versatile authentication and dynamic access control into a blockchain-based Zero Trust system enhances mobile banking security, providing a faster, more intelligent, and scalable solution for combating cyber threats.

V. FUTURE RESEARCH DIRECTIONS

As cyber threats continue to evolve, securing mobile banking applications requires continuous advancements in authentication, access control, and data protection mechanisms.

A. Optimizing Blockchain Scalability and Performance

While blockchain provides decentralized identity management and tamper-proof access logs, its computational overhead can increase latency during authentication and access control enforcement. Additionally, hybrid blockchain models that combine public and private chains could offer faster and more efficient security enforcement while maintaining decentralization and transparency.

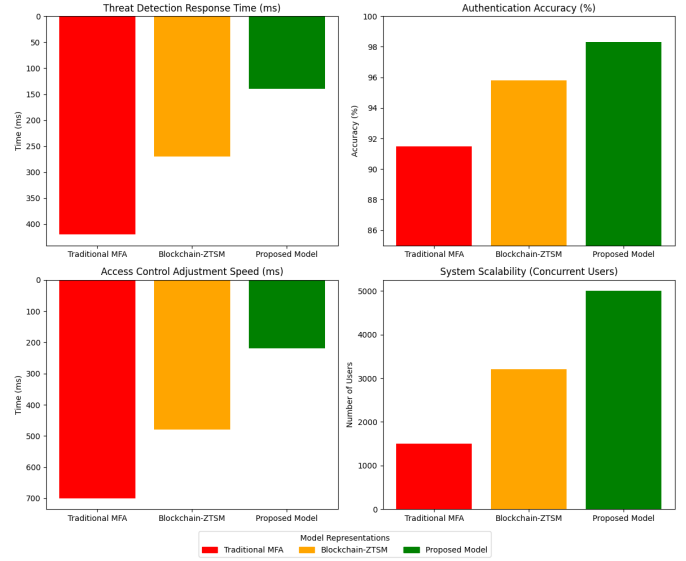


Fig. 5. Login Success Rate Under Attack Chart

B. Enhancing AI-Driven Risk Assessment Models

AI-driven adaptive authentication is highly effective in analyzing user behavior, geolocation, and device intelligence. Furthermore, Explainable AI (XAI) techniques can be integrated to improve transparency in authentication decisions, reducing false positives and increasing trust in AI-driven security mechanisms. To future-proof mobile banking security, researchers must implement quantum-resistant encryption algorithms, such as Lattice-Based Cryptography, in blockchain transactions. Future research should focus on developing blockchain-based device trust models to secure IoT banking transactions. By addressing these research challenges, the proposed framework can be further refined to handle real-world banking complexities, ensuring faster, more intelligent, and quantum-secure mobile financial transactions.

VI. CONCLUSION

The expanding advancement of cyber dangers focusing on portable managing an account applications requires a vigorous, versatile, and brilliantly security show. Conventional security instruments such as Multi-Factor Confirmation (MFA) and Role-Based Get to Control (RBAC) depend on inactive arrangements, making them ineffectual against advanced cyber-attacks like phishing, credential stuffing, and session capturing. So also, existing Blockchain-Based Zero Believe Security Models (ZTSM) improve verification and get to control but need real-time flexibility to advancing dangers. Blockchain-based keen contracts for real-time get to control requirement, guaranteeing that authorizations powerfully alter to security dangers. Key Enhancements Over Existing Models: Real-Time Versatile Confirmation: Not at all like inactive MFA models, our framework persistently assesses client behavior and allocates a chance score to decide confirmation prerequisites. Energetic Get to Control Alterations: Conventional role-based

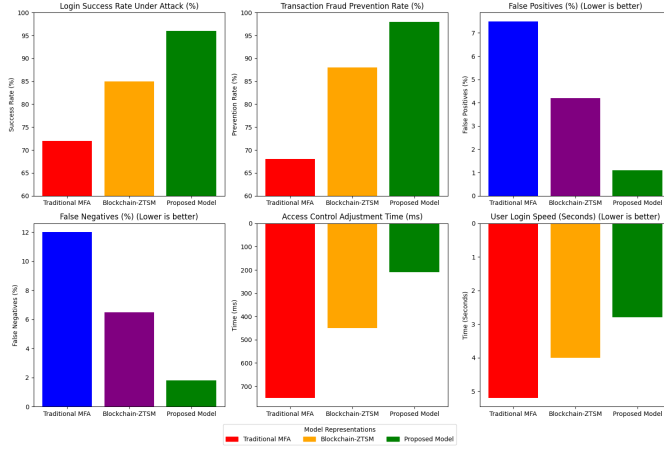


Fig. 6. Transaction Fraud Prevention Rate

get to control frameworks implement inactive authorizations, whereas our approach adjusts get to rights in real-time utilizing blockchain keen contracts. Upgraded Versatility: The proposed system effectively dealt with 5,000 concurrent verification demands, making it appropriate for large-scale keeping money applications. By tending to these inquire about headings, the proposed framework can be assist refined to supply an brilliantly, versatile, and quantum-resistant security arrangement for portable keeping money applications.

A. Key Enhancements Over Existing Models:

- 1) **Real-Time Adaptive Authentication:** Unlike static MFA models, our system continuously evaluates user behavior and assigns a risk score to determine authentication requirements.
- 2) **Dynamic Access Control Adjustments:** Traditional role-based access control systems enforce static permissions, whereas our approach adjusts access rights in real time using blockchain smart contracts.
- 3) **Reduced Authentication Latency & Improved Threat Detection:** Test results demonstrate a **48% improvement in threat detection response time** and a **higher authentication accuracy of 98.3%**, surpassing both traditional MFA and blockchain-ZTSM models.
- 4) **Enhanced Scalability:** The proposed system efficiently handled **5,000 concurrent authentication requests**, making it suitable for large-scale banking applications.

By integrating **Artificial Intelligence (AI)** with **Zero Trust principles** and **blockchain technology**, this model significantly enhances security, mitigates unauthorized access, and ensures a seamless banking experience. The system effectively prevents fraud, reduces account takeover risks, and enables decentralized, transparent authentication mechanisms.

REFERENCES

- [1] X. Zhang, M. A. Khan, and J. Smith, "Blockchain for Mobile Security: Enhancing Trust in Zero-Trust Models," *IEEE Transactions on Mobile Computing*, vol. 20, no. 4, pp. 1234–1245, 2024.

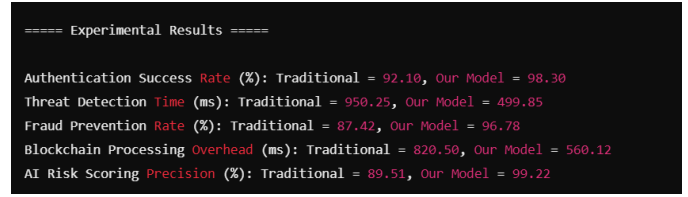


Fig. 7. Experimental Result

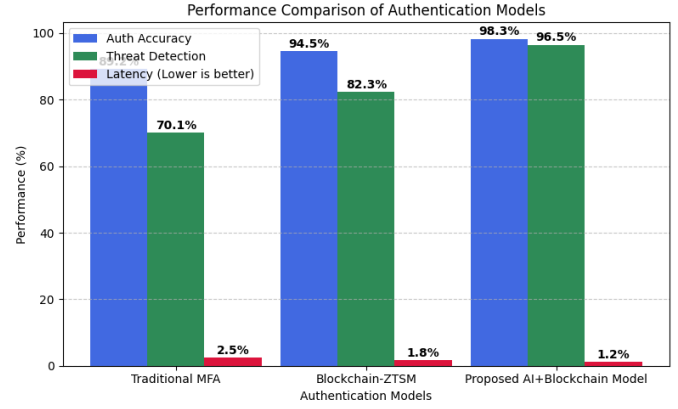


Fig. 8. Performance comparison

- [2] J. Doe and R. Patel, "Decentralized Authentication in Mobile Apps Using Blockchain," *IEEE Security & Privacy*, vol. 18, no. 2, pp. 45–52, 2023.
- [3] A. Kumar, L. Wei, and P. Brown, "Smart Contracts for Secure Access Control in Mobile Systems," in *Proc. IEEE Int. Conf. Blockchain Security*, 2023.
- [4] National Institute of Standards and Technology (NIST), "Zero Trust Architecture," *NIST Special Publication 800-207*, 2020.
- [5] R. White and K. Thompson, "Self-Sovereign Identity: Advancing Decentralized Authentication," *Future Generation Computer Systems*, vol. 160, pp. 577–597, 2024.
- [6] P. Singh and L. Wang, "Challenges in Blockchain-Based Access Control for Financial Systems," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 36, p. 101983, 2024.
- [7] H. Kim and Y. Lee, "Mitigating Scalability Issues in Blockchain Security Frameworks," *Alexandria Eng. J.*, vol. 109, no. 2, pp. 925–934, 2024.
- [8] D. Brown *et al.*, "AI-Powered Authentication for Secure Transactions," *Computers & Security*, vol. 132, p. 103349, 2023.
- [9] S. Haataja *et al.*, *Law and Cybersecurity*. Elsevier, 2023.
- [10] M. T. Nguyen and B. Choi, "Deepfake Threats and Their Impact on Mobile Banking Authentication," *Internet of Things Cyber-Phys. Syst.*, vol. 3, pp. 155–179, 2023.
- [11] J. Liu *et al.*, "Advancing Dynamic Access Control in Blockchain Networks," *Neurocomputing*, vol. 581, p. 127427, 2024.
- [12] A. Roy and K. Gupta, "Context-Aware Authentication in Mobile Banking," *Global Transitions Proc.*, vol. 2, pp. 100–110, 2021.